ModelIt!

Student Activity Pack

Is Fast Fashion Destroying the Planet?

Modeling the Environmental Life Cycle of Clothing

Grade 11L1 | NGSS HS-ESS3-2, HS-ETS1-3 | Energy and Matter

STUDENT NAME _____ DATE _____

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1 Pre-Assessment

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What Do You Know? Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

1. **How many new pieces of clothing do you think you buy in a typical year? How many times do you wear each item before you stop wearing it?**

2. **What do you think happens to clothing after you throw it away or donate it to a thrift store?**

3. **Do you think a $5 t-shirt and a $30 t-shirt have the same environmental impact? Why or why not?**

4. **What resources do you think are needed to make a single cotton t-shirt — water, energy, chemicals, land?**

2 Scoring Rubric

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Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

3 Component Analysis

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What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

Sort These Components

Production Volume | Material Type | Garment Lifespan | Water Consumption | Carbon Emissions | Textile Waste Generated

EXTERNAL	INTERNAL

? Explain your reasoning: Why did you sort each component the way you did?

4 Model Recording Page

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Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

? Explain ONE relationship in your model. What happens when one component changes?

5 Simulation Observations

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Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

1 Scenario 1: Fast Fashion Default

Production: Maximum | Material: Polyester | Lifespan: 7 wears

MY PREDICTION

WHAT I OBSERVED

2 Scenario 2: Sustainable Wardrobe

Production: -50% | Material: Organic cotton/recycled | Lifespan: 100 wears

MY PREDICTION

WHAT I OBSERVED

3 Scenario 3: Circular Fashion System

Production: Moderate | Fiber recycling: 70% recovery

MY PREDICTION

WHAT I OBSERVED

6 Research & Extend

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Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

Possible New Components

Labor Conditions — The working conditions of garment factory workers, predominantly in Bangladesh, Vietnam, and Cambodia — fast fashion's low prices are enabled by wages as low as $2-3/day in factories with poor safety standards, connecting environmental exploitation to human exploitation

Microplastic Accumulation — The progressive buildup of synthetic microfibers in oceans, soil, drinking water, and human tissue — an estimated 14 million tons of microplastic sit on the ocean floor, and microplastics have been found in human blood, lungs, and placenta

Consumer Culture Pressure — The social media-driven pressure to constantly update wardrobes — algorithms that promote 'haul' videos and trend cycles accelerate consumption, with the average number of times a garment is worn declining 36% in the last 15 years

My Extension

A Which new component would you add to the model?

B How would it connect to the existing components? (Draw or describe)

C What evidence or reasoning supports your connection?

7 Reflection Questions

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The Big Question

If a $5 t-shirt costs the planet 2,700 liters of water, 10 kilograms of CO_2, and microplastic pollution that lasts centuries — who really pays the price? Use evidence from your simulation to support your answers.

1. **How does your model reveal the hidden environmental costs that are not reflected in the purchase price of fast fashion clothing?**

2. **What surprised you most about the relationship between garment lifespan and total environmental impact?**

3. **Why do you think consumers continue to buy fast fashion despite growing awareness of its environmental consequences?**

4. **How could your model be used to inform policy — such as environmental labeling requirements or extended producer responsibility laws?**

5. **What limitations does your model have in capturing the full impact of the fashion industry, including labor conditions and social effects?**

8 STEM Challenge: Design a Sustainable Fashion Brand

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YOUR ENGINEERING MISSION

Design a clothing brand that produces fashionable garments while reducing environmental impact by at least 70% compared to fast fashion benchmarks across water, carbon, and waste metrics.

THE SCENARIO

A venture capital firm wants to invest in a sustainable fashion startup and has hired your environmental science team to design the brand's environmental strategy. The brand must produce affordable, stylish clothing while reducing per-garment water consumption by 70%, carbon emissions by 70%, and textile waste by 80% compared to fast fashion industry averages. Your design must specify material choices, production processes, garment durability targets, and end-of-life strategies.

Engineering Design Thinking

1. WHAT SPECIFIC FIBERS WILL YOU USE AND WHAT ARE THE ENVIRONMENTAL TRADE-OFFS OF EACH COMPARED TO CONVENTIONAL POLYESTER AND COTTON?

2. HOW WILL YOU KEEP PRICES ACCESSIBLE TO TEENAGERS WHILE ABSORBING THE HIGHER COSTS OF SUSTAINABLE MATERIALS AND ETHICAL PRODUCTION?

3. WHAT TAKE-BACK OR RECYCLING PROGRAM WILL YOU IMPLEMENT, AND WHAT PERCENTAGE OF SOLD GARMENTS DOES YOUR MODEL PREDICT WILL ACTUALLY BE RETURNED?

4. HOW WILL YOU COMMUNICATE THE TRUE COST-PER-WEAR TO CONSUMERS TO SHIFT PURCHASING BEHAVIOR AWAY FROM DISPOSABLE FASHION?

My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

! Explain your design using scientific evidence from the simulation:

Real Career Connection

Sustainable Fashion Designers create clothing using environmentally responsible materials and processes, earning $50,000–$110,000/year at eco-conscious brands. Life Cycle Assessment Analysts quantify the environmental impact of products from raw materials to disposal, earning $65,000–$120,000/year at consulting firms, manufacturers, and government agencies. Textile Engineers who develop sustainable fibers and recycling technologies earn $70,000–$130,000/year.